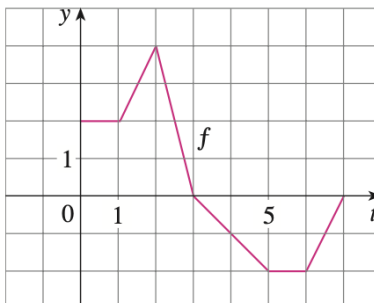


Tutorial - Week 8

TG 11 - TG 12

1. Let $g(x) = \int_0^x f(t) dt$, where f is the function whose graph is shown.



- (a) Evaluate $g(0)$, $g(1)$, $g(2)$, $g(3)$ and $g(6)$.
- (b) On what interval is g increasing?
- (c) Where does g have a maximum value?
- (d) Sketch the graph of g .

2. On which interval is the curve

$$y = \int_0^x \frac{1-t^2}{1+t^2} dt$$

concave downward?

3. (a) Let $g: (c, d) \rightarrow [a, b]$ be a differentiable function, and $f: [a, b] \rightarrow \mathbb{R}$ be a continuous function. For $x \in [a, b]$, define

$$G(x) = \int_a^{g(x)} f(t) dt.$$

Show that

$$G'(x) = f(g(x))g'(x).$$

- (b) Let $u, v: (c, d) \rightarrow [a, b]$ be two differentiable functions, and $f: [a, b] \rightarrow \mathbb{R}$ a continuous function. Find a formula for

$$\frac{d}{dx} \int_{u(x)}^{v(x)} f(t) dt.$$

4. Find the derivative of the following functions.

(a) $g(x) = \int_0^x e^{-t^2} dt.$

(f) $g(x) = \int_{3x}^{e^{3x}} \ln(t) dt.$

(b) $g(x) = \int_x^1 \cos^2(3t-2) \ln t dt.$

(g) $g(x) = \int_0^{\cosh(x^2-x)} 1 dt.$

(c) $g(x) = \int_0^{3x} \frac{1}{\sqrt{4-t^4}} dt.$

(h) $g(x) = \int_{-2x}^{2x} \frac{1}{(t-1)^2 + 1} dt.$

(d) $g(x) = \int_4^{2x^2} \frac{1}{2t-2} dt.$

(i) $g(x) = \int_{-x}^x \sin t dt.$

(e) $g(x) = \int_{x^2-1}^0 (t^3 + t^2 + t + 1) dt.$

(j) $g(x) = \int_{u(x^2)}^{v(x^3)} f(t) dt$, with u and v two differentiable functions and f continuous.

5. Apply twice the Fundamental Theorem of Calculus to compute the 2nd derivative of the integral function

$$\frac{d^2}{dx^2} \left[\int_0^x \left(\int_0^{\sqrt{1+u^4}} (\sinh(t) + t) dt \right) du \right].$$

6. Find a function f and a number a such that

$$\int_a^x \frac{f(t)}{t^2} dt + 6 = 2\sqrt{x}.$$

7. Find the following integrals.

(a) $\int \frac{x^3}{(x^2 + 3)^3} dx.$

(e) $\int \frac{\sin^3 x}{\sqrt{\cos x}} dx.$

(b) $\int \frac{\ln x}{x(1 + (\ln x)^2)^2} dx.$

(f) $\int \frac{\sqrt{x^2 + 1}}{x^2} dx.$

(c) $\int \frac{1}{x\sqrt{\ln x^2 + 1}} dx.$

(g) $\int \frac{\sin^{-1}(x)}{\sqrt{1 - x^2}} dx.$

(d) $\int \frac{e^x}{\sqrt{1 - e^{2x}}} dx.$

(h) $\int \frac{\tan^{-1}(x)}{1 + x^2} dx.$

8. Find all f such that $f(x) = f'(x)(4x + 9)^5$ and $f(-2) = 5$.
9. Find the area enclosed by the graphs $(x - 2)^2 - 2y = -2$, $y = x + 2$, $x = -2$, and $x = 6$.
10. Find the area of the region enclosed by the curves $f(x) = \sqrt{x}$, $y = 3$, and the tangent line to f at $x = 1$.
11. Find the area of the region enclosed by the curves $y^2 = 2x$, $3y = -2x + 10$, and $y = 4$.
12. Find the volume of the solid of revolution obtained by rotating the region from the previous exercise around the x -axis.
13. Consider the region $\mathcal{R}$ enclosed by the curves $y = \sqrt{x}$ and $y = x$. Find the volume of the resulting solid if
- (a) The region $\mathcal{R}$ is rotated around the x -axis.
 - (b) The region $\mathcal{R}$ is rotated around the line $y = \frac{3}{2}$.